

Economic evaluation and analysis of the German and French grids environmental footprint within the scope of production and implications of nuclear power plant closures

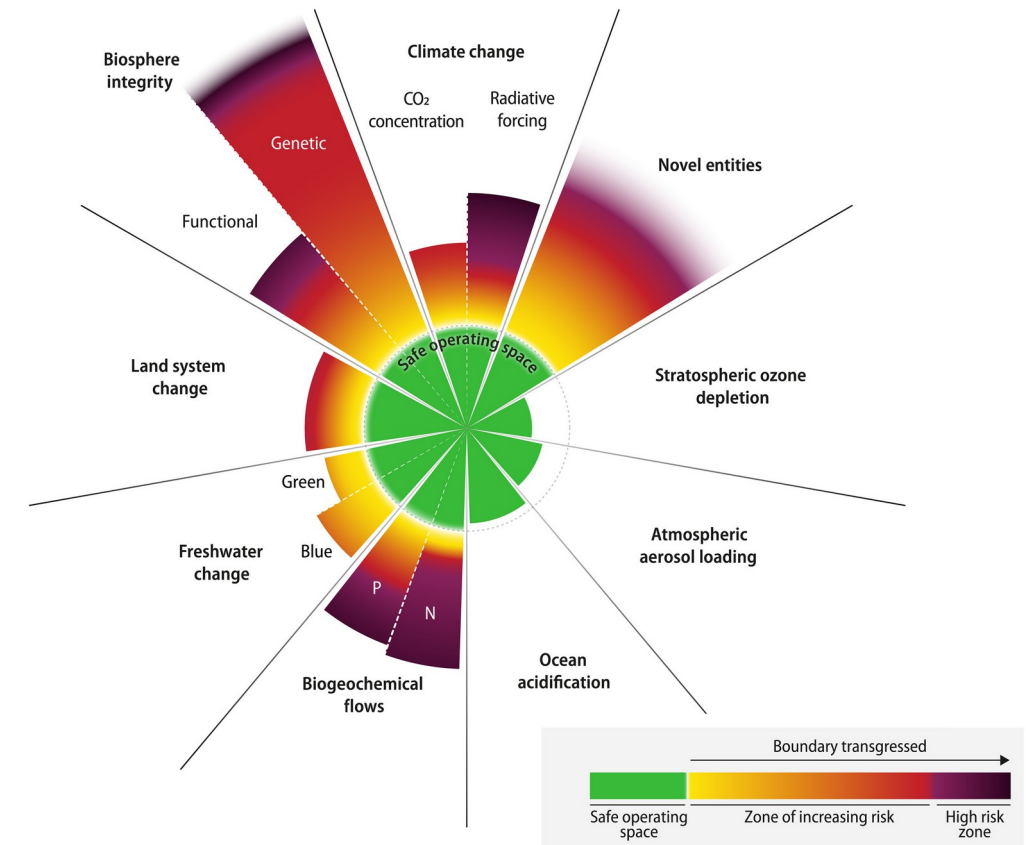
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Climate change is unfortunately not our only emergency regarding the integrity of the biosphere

“Earth is now well outside of the safe operating space for humanity” (Richardson et al, Science Advances, Sept. 2023)

The planetary boundaries framework quickly became a pillar of Earth system science with a growing influence

➔ There is an urgent need to look beyond greenhouse gases emission to further rationalize our choice energy policy choices (RTE, Futures énergétiques, 2021)



Purposes of this work

1. Produce a yardstick to compare impact of various power generation technologies beyond greenhouse gases in a way that is both comprehensive and standard
2. Give a (monetary) measure of the decrease/increase of nuclear output in different electricity mix
3. Try to impulse a dynamic for this kind of work with a publication (hopefully) in Energy Policy

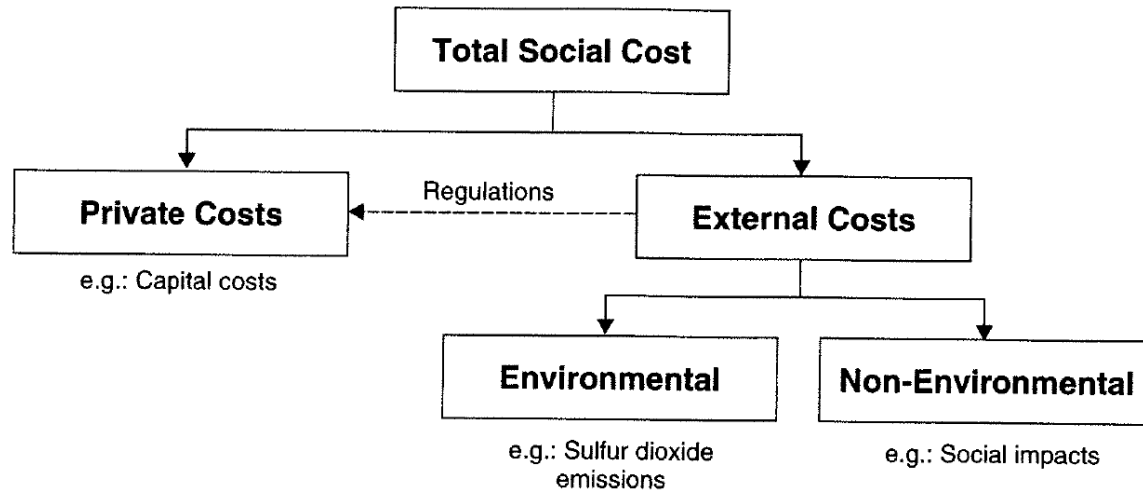
Life Cycle Assessment (LCA)

➔ Objectivize impacts over the entire life cycle ('from cradle to grave') of service/goods production

Extensive literature on the impact per kWh of electricity for several technologies, but:

1. Largely focused on the global warming indicator (*Barros, et al, 2020*) ➔ in this study, we use the environmental footprint framework, 16 indicators)
2. Difficulty of aggregating impact measures into complex units ➔ weighting method single score (much debated, "half science/half art"). One way to do so is **monetary valuation**

Nature of external environmental costs



Externality = effect (positive or negative) not mediated by an economic institution

Regulatory mechanism = market-based solution (EUTS) or not (carbon tax)



Double counting = internalizing "external costs" that are already partly internalized (carbon, ionizing radiation, etc.).

How can we assess these external environmental costs?

Two main methods: damage cost or abatement cost

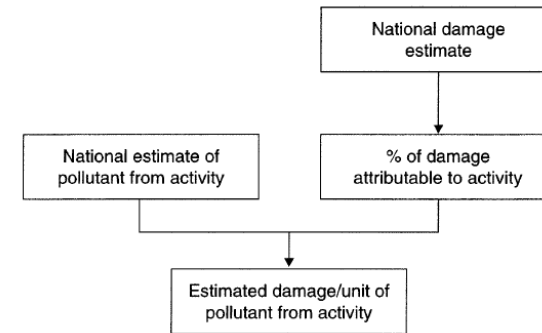
Abatement cost = Cost of regulations to "mitigate" and control the damage generated by the externality.



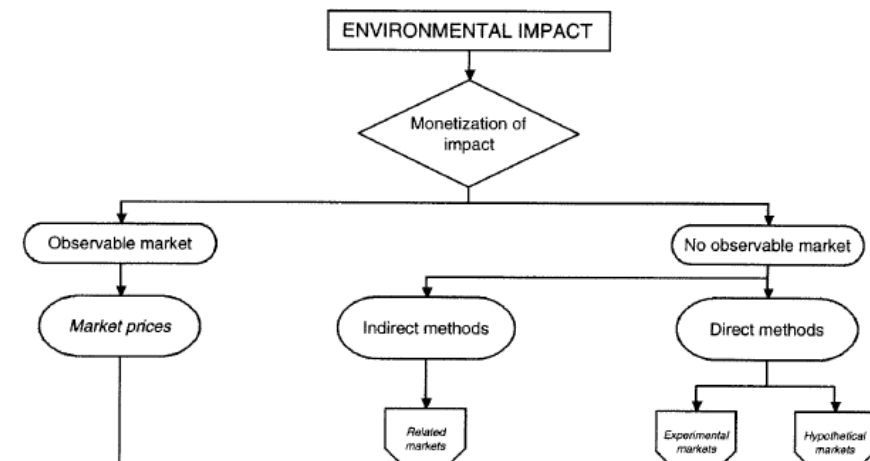
→ requires decision-makers to make the right decisions

Cost of damage (2 approaches):

- ❖ Bottom-up
- ❖ Top-down



Top-down approach



Bottom-up approaches (indirect methods: **revealed preferences** / direct methods: **contingent valuation**)

Many critics ...

- Dependence on social context
 - Incorporating implicit ethical standards into valuation methods
 - Extending the domain of "commensurable" to none-socialized Nature
 - Arrow's theorem on the impossibility of aggregating individual preferences
 - WTP/MTA difference of several orders of magnitude
- Etc. etc. (critical studies discussed in the paper)

...but...

- 1) Performing an LCA implicitly recognizes the weighting of impacts
- 2) Implicitly/De facto, we are already placing values on these "incommensurable" things → benefit for these things to objectify this value
- 3) Already used at the European level (« Directive Cadre européenne sur l'eau »)
- 4) Using these methods doesn't prevent us from discussing the methodological aspects (practitioners are in favor of using these methods).
- 5) Allows you to draw salient conclusions
- 6) Do not confuse the value of X with the monetary valuation of X
- 7) Objectivize the values we assign and give less power to politicians

Materials used in the study

LCA environmental footprint of technologies → UNECE
 (+ comparisons with existing literature, a dozen studies)
 European LCA



No resources to do our own LCA
 → production hypothesis impacting LCA results

Monetary valuation → Amadei et. al, 2021
Extensive and Systematic literature review of 20+ studies
 (Triconomics, DE Delft etc.)



1. For same methods, large differences (e.g. ionizing radiation, coefficient of variation > 70%) → **median**
 2. **Which discounting ?**

TABLE 1. LIST OF MONERATY VALUATION COEFFICIENTS IN €2019.

Method	Midpoint impact indicator	Median value	Number of valuations included
Damage cost	Climate change (€/kg eq. CO ₂)	1,86E-01	4
Damage cost	Acidification (€/mol eq. H ⁺)	1,03E+00	7
Damage cost	Freshwater ecotoxicity (€/CTU)	3,88E-05	2
Damage cost	Freshwater eutrophication (€/kg eq. P)	3,83E+00	6
Damage cost	Marine eutrophication (€/kg eq. N)	1,61E+00	4
Damage cost	Terrestrial eutrophication (€/mol eq. N)	1,12E-01	3
Damage cost	Carcinogenic effects (€/CTUh)	5,75E+05	5
Damage cost	Ionizing radiation (€/kg eq. U235)	1,22E-03	5
Damage cost	Non-carcinogenic effects (€/CTUh)	5,75E+05	5
Damage cost	Ozone layer depletion (€/kg eq. CFC-11)	4,61E+01	5
Damage cost	Photochemical ozone creation (€/kg eq. NMVOC)	1,18E+00	7
Damage cost	Respiratory effects, inorganics (€/disease)	7,98E+05	1
Market price	Particulate matter (€/kg eq. PM _{2.5})	3,36E+01	4
Market price	Dissipated water (€/m ³)	1,43E-02	2
Damage cost	Fossils depletion (€/MJ)	1,03E-02	8
Damage cost	Land use (€/pts)	2,03E-03	2
Market price	Mineral and metals depletion (€/kg eq. Sb)	1,92E+01	6

LCA comparison UNECE with EDF

TABLE 2. COMPARISON BETWEEN TWO LCA OF ONE NUCLEAR KWH GENERATED OVER SEVERAL IMPACT MIDPOINT INDICATORS

(Indication: CC: climate change, O: ozone layer depletion, IR: ionizing radiation, POC: photochemical ozone creation, A: acidification, FE: freshwater eutrophication, TE: terrestrial eutrophication, ME: marine eutrophication, MMD: minerals and metals depletion)

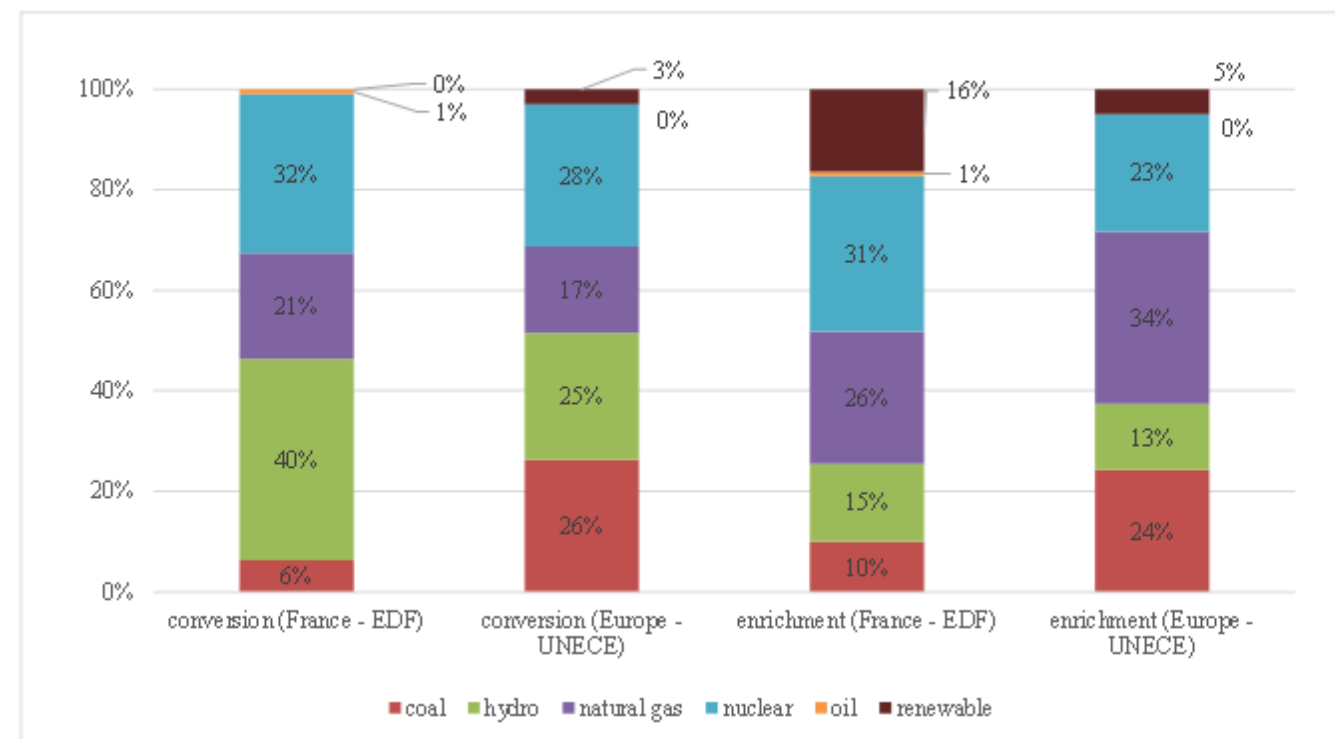
Indicator (emissions per kWh produced)	CC (kg eq. CO2)	O (kg eq. CFC-11)	IR (kg eq. U235)	POC (kg eq. NMVOC)	A (mol eq. H+)	FE (kg eq. P)	TE (mol eq. N)	ME (kg eq. N)	MMD (kg eq. Sb)
[44]	3,7e-3	4,7e-10	6,80e-1	2,10e-5	3,40e-5	1,90e-7	7,10e-5	1,50e-5	4,80e-6
[42]	5,29e-3	4,62e-10	4,75e-1	2,65e-5	4,28e-5	6,45e-6	9,70e-5	8,20e-5	3,33e-7

LCA comparison UNECE with EDF

Etape / indicateur	Mines	Combustible	Production	Traitement CU	Déchets
Changement climatique					
Appauvrissement Ozone					
Particules					
Ozone photochimique					
Acidification					
Eutrophisation Terrestre					
Eutrophisation Eau douce					
Eutrophisation Marine					
Radiations ionisantes					
Epuisement Ressources					

Tableau 3 : vision synthétique de l'ACV du kWh nucléaire EDF SA

FIG. 1. Comparison of electricity mixes for conversion and enrichment used in [42] and [44] (author's calculations based on various sources, including [44] for supply portfolio data for France, and national statistics for electricity mix)

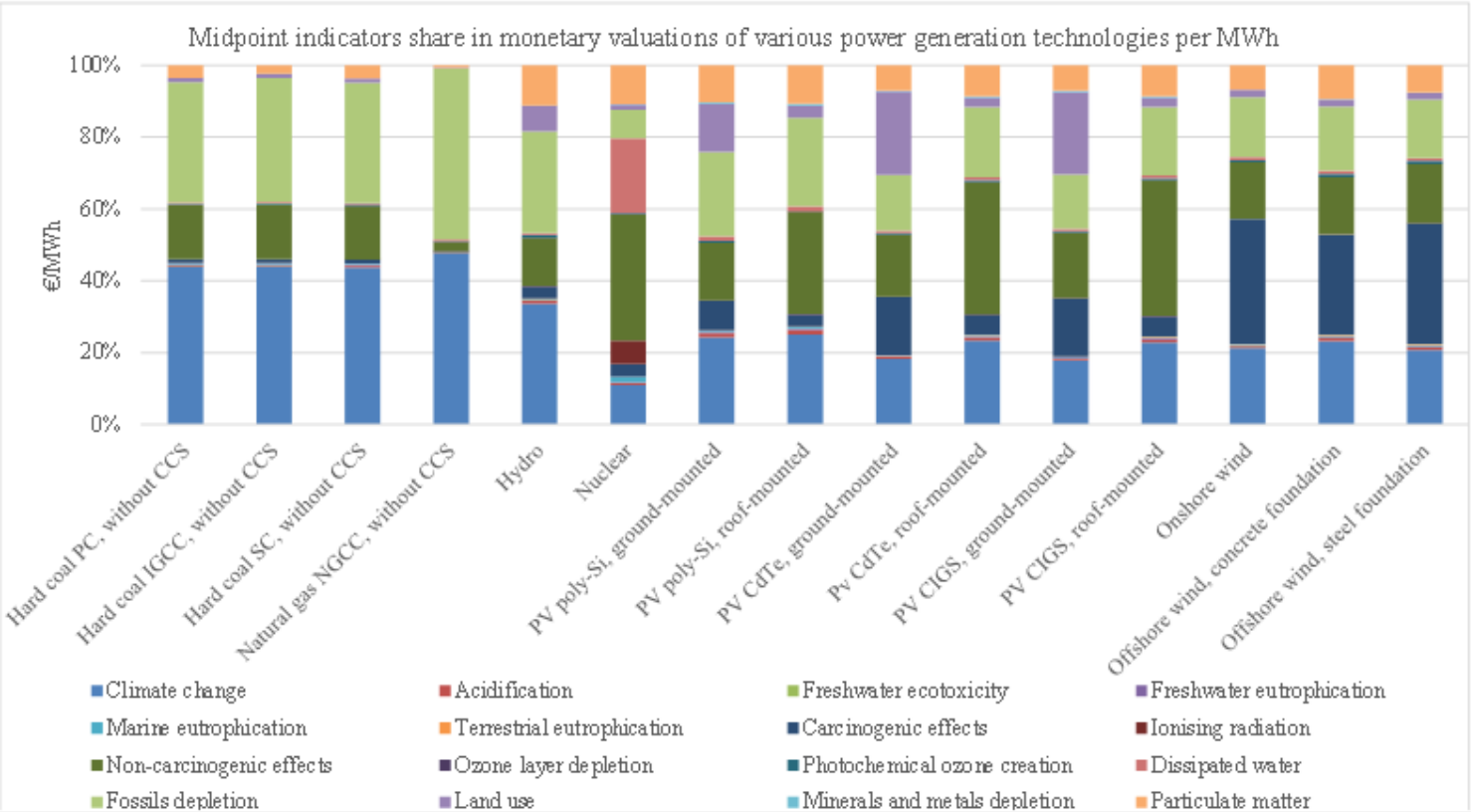


Possible explanations: no fuel recycling in UNECE, higher carbon mix for conversion/enrichment in UNECE ... (?)

→ Sensitivity study carried out on environmental costs: low impact (5-15%)

Environmental cost by technology

FIG. 2. Midpoint indicators share in monetary valuations of various power generation technologies per MWh.
NB: NGCC = natural gas combined cycle



Power generation technology	Environmental cost (€/MWh)
Hydro	6
Nuclear	9
Onshore wind	11
PV CdTe	12
Offshore wind	12
PV CdTe, ground-mounted	12
PV poly-Si, roof-mounted	28
PV poly-Si, ground-mounted	28
Nature Gas Combined Cycle	170
Oil	250
Hard coal (and brown coal)	400

What's important is not the precise value, given the uncertainties involved. It's the **orders of magnitude** and the **most significant impact items** that matter.

➔ **Orders of magnitude:** Net difference between carbon-based energies (> €100/MWh) and renewable energies, hydro and nuclear (< €50/MWh)

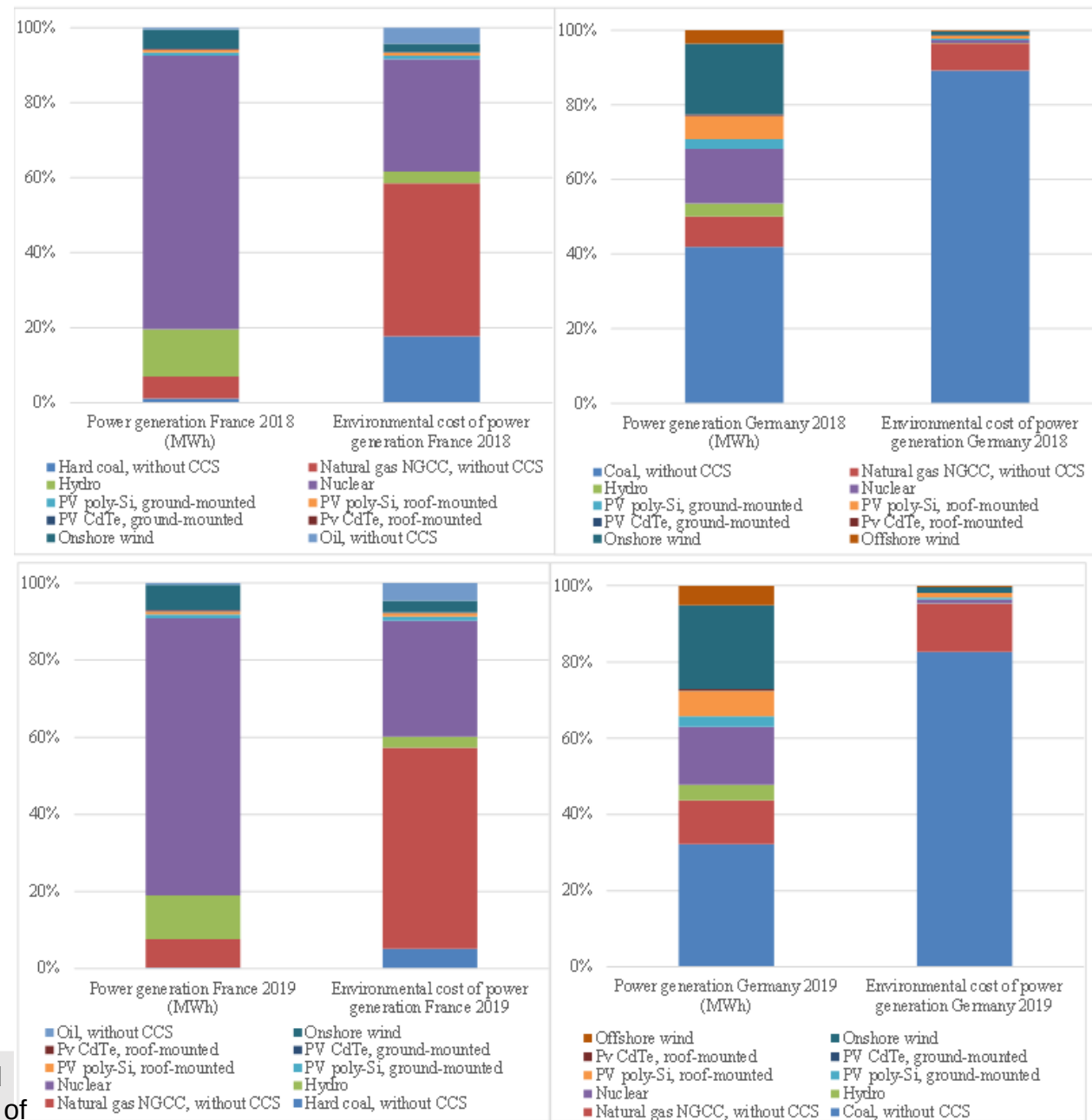
➔ **Impact items:** On nuclear power, "water use » (water consumption vs. water withdrawal discussion) and the rationale behind a market price for water for this item), whereas hydro...
Land use / Ionizing radiation of course / etc.

Electricity mix vs. contribution to environmental costs

→ In France, **gas** contributes ~5% of production vs. ~40% of the environmental cost. For **nuclear power**, the contribution to production is ~75% vs. ~30% of the unit environmental cost.

→ Contrast between technologies « *environmentally beneficial* " in the **current power system configuration** vs. « non- ... »...

FIG. 3. Share in power generation and contribution of generation technologies to environmental cost in France (left) and Germany (right) in 2018 (above) and in 2019 (below).



Average environmental cost of domestic production

	2018	2019
France	23.6 €/MWh	23.1 €/MWh
Germany	188 €/MWh	155 €/MWh

Power generation technology	Environmental cost (€/MWh)
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PV poly-Si, ground-mounted	28
Nature Gas Combined Cycle	170
Oil	250
Hard coal (and brown coal)	400

If costs were internalized, *"non-environmentally beneficial"* technologies would leave the market:

- ➔ *In France, this mainly concerns gas*
- ➔ *In Germany, this currently only applies to coal.*

Work in progress

Econometric model (Woo et al. 2011a; Woo et al. 2013; Woo et al. 2014; Kennedy 2016).

The idea is to answer the question: what is the environmental impact (quantified in monetary terms) of reducing nuclear output of 1 unit ?

$$P_{kht} = \alpha_{kht} + \underbrace{\sum_{i=1}^N \gamma_{kht}^i C_{kht}^i}_{(i)} + \underbrace{\sum_{i=1}^N \widehat{\gamma}_{kht}^i C_{kht}^i}_{(ii)} \underbrace{PH_{kht}}_{(iii)} + \underbrace{\beta_k Q_{kht}}_{(iv)} + \underbrace{\widehat{\beta}_{kht} Q_{kht}}_{(v)} \underbrace{PH_{kht}}_{(vi)} + \underbrace{\psi_{kht} N_{kht}}_{(v)} + \underbrace{\widehat{\psi}_{kht} N_{kht}}_{(vi)} \underbrace{PH_{kht}}_{(vi)} + \underbrace{\epsilon_{kht}}_{(vii)}$$

P = environmental cost; k {Germany, France, other countries ?} ; h hours of the day; t day of the week

- i. Hourly domestic electricity production for technology i
- ii. Hourly domestic electricity production of technology i with $PH = 1$ for peak hours (EEX definition)
- iii. Household consumption
- iv. Household consumption with $PH = 1$...
- v. Hourly electricity production Fessenheim or Philippsburg (either available hourly times, or pro rata to domestic nuclear production)
- vi. ... with $PH = 1$...
- vii. Error terms: either ARCH model (time-dependent error variance to account for Renewables growth) or SARIMA model (error seasonality)

Many questions left open

- Externality (political cost) of an accident (case of Germany)
- The underlying assumptions of monetary valuations (taking time into account in valuations, etc.) cannot be ignored
- Is taking a single monetary evaluation value the right choice (order-of-magnitude approach, min max, perhaps even MCMC method to derive an environmental cost, etc.)?
- With regard to nuclear power: the subject of **water** and the subject of **accidents** : are accidents taken into account across the board in different impact categories: ionizing radiation/ carcinogenic effects etc. or not?
- Does it make sense to look at the externalities of a fleet in transition vs. a stabilized fleet (2050)?
- We also need to look at the impact of the network, whose topology and infrastructure differ greatly between the extreme scenarios on the role of nuclear power.

Key takeaways

- Monetary valuations could be useful for public decision-makers
- Research is needed in monetary valuation methodologies and should be encourage
- Nuclear is among the least environmentally harmful technology
- Clear gap between carbon-based and non-carbon-based technologies → robust conclusion to be drawn here beyond the methodological difficulties
- Carbon-based power generation technologies are the most environmentally costly → **further incentivize to phase out of fossil fuels technologies**
- Major shifts in EOM can be expected if environmental costs were internalized



Thank you!